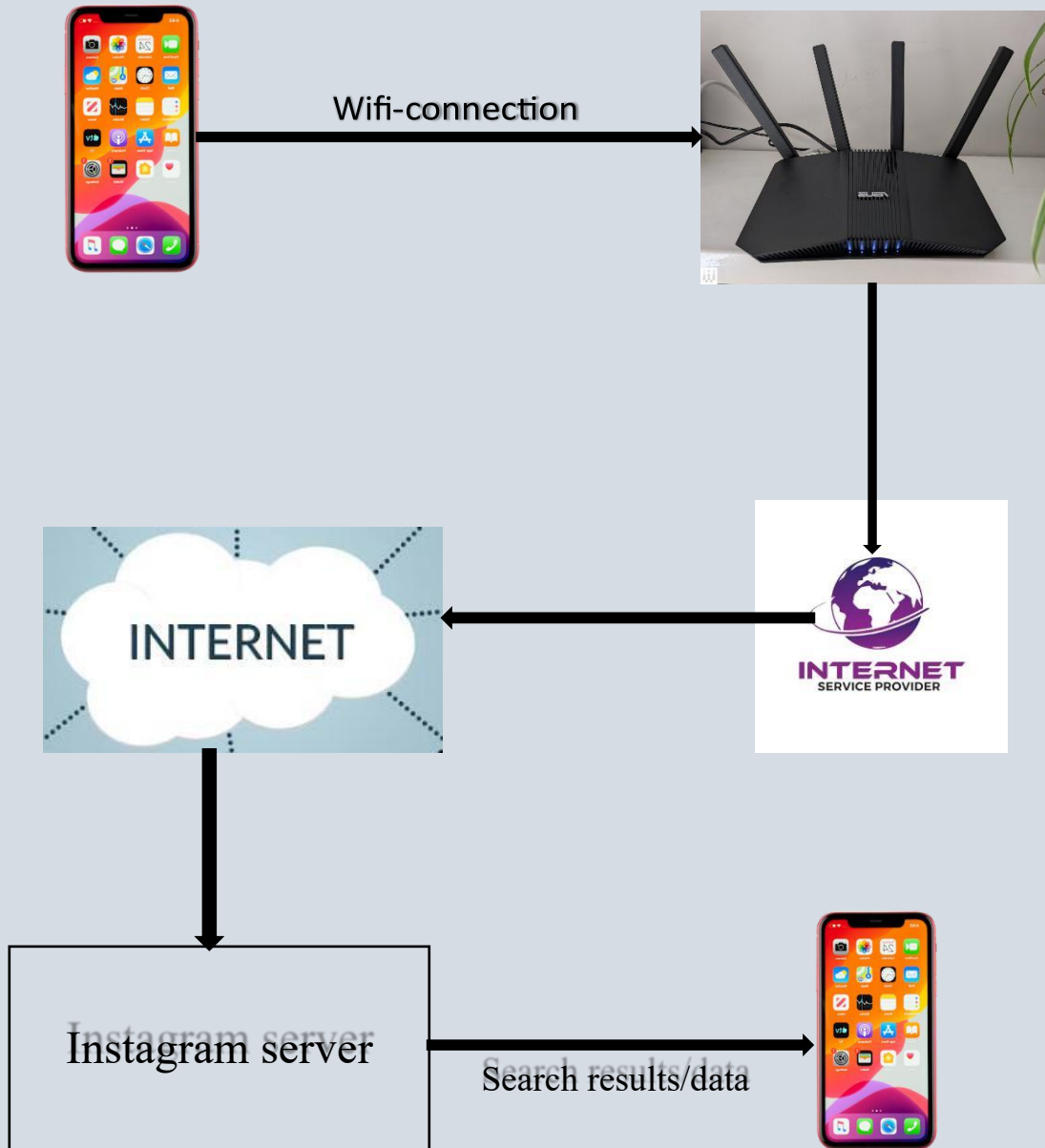


## *Introduction to networking*

1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



**2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).**



| Device       | Type of Network                 | Example Use                                                  |
|--------------|---------------------------------|--------------------------------------------------------------|
| Smartphone   | WAN (Mobile Data) / LAN (Wi-Fi) | Browsing the internet and using social media.                |
| Laptop       | LAN (Wi-Fi or Ethernet)         | Studying, browsing, and online meetings.                     |
| Wi-Fi Router | LAN & WAN                       | Connects home devices to the internet.                       |
| Smart TV     | LAN (Wi-Fi)                     | Streaming YouTube, Netflix, and other online content.        |
| Smartwatch   | PAN (Bluetooth)                 | Connects to a smartphone to receive calls and notifications. |

**3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.**



| Internet                                                   | Intranet                                                                         |
|------------------------------------------------------------|----------------------------------------------------------------------------------|
| The Internet is a global network that anyone can access.   | An Intranet is a private network used only within an organization.               |
| It is used for websites, email, YouTube, and social media. | It is used to share files, notices, and information within a college or company. |
| Anyone with an internet connection can use it.             | Only authorized users can access it.                                             |

**4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.**  
***Hint: Mention the devices and networks involved, and what happens when you hit 'search'.***



**Example: Zomato App Search**

When I search for a restaurant in the Zomato app, my mobile sends the request through the Wi-Fi/mobile network to the Internet Service Provider (ISP).

The request travels through routers and reaches Zomato's servers, where the search data is processed.

Zomato's server finds the required restaurant information from its database and sends the response back through the network.

My mobile receives the data and displays the restaurant list on the app screen.

The communication happens using networking protocols like HTTPS to keep the data secure.